Supplementary Information for

"Systems-level Modeling of Cell Migration using Spatially-Resolved Single-Cell Data"

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Supplementary Note S1. Conceptual motivation

The computational analyses presented in this study are conceptually motivated by established continuum and compartmental modeling frameworks that have been widely used to describe cell population dynamics. These equations are introduced solely to provide theoretical context for the descriptive metrics proposed here and are not part of the computational pipeline. No parameter of either equation is estimated, optimized, or fitted, and neither equation is used to perform spatial deconvolution, cell assignment, or temporal inference. Spatiotemporal cell-type redistribution can be conceptually related to a reaction-diffusion process:

$$\frac{\partial c(x,y,t)}{\partial t} = D\nabla^2 c(x,y,t) + f(c) \qquad (1)$$

where $c(x,y,t)$ denotes the inferred abundance of a given cell type at spatial location (x,y) and time t, D denotes an effective dispersion coefficient, and $f(c)$ represents local biological processes, including proliferation, cell death, and microenvironmental interactions. Inferred cell-state dynamics can similarly be related to a compartmental transition model:

$$\frac{\partial p_i(t)}{\partial t} = \sum_{j \neq i} k_{ji} p_j(t) - \sum_{j \neq i} k_{ij} p_i(t) \qquad (2)$$

where $p_i(t)$ represents the abundance of a cell type i at time t, and k_{ij} represents the inferred transition rate from cell type i to cell type j.

These equations are intended only to illustrate biological processes that may contribute to the temporal patterns captured by the proposed descriptive framework. They do not constitute mathematical formulations of the implemented algorithm. The computational pipeline operates independently through CARD-based spatial deconvolution, CARD_SCMapping-based spatial assignment, and quantitative characterization of spatiotemporal cell-type redistribution using Centroid Shift, Abundance-Weighted Centroid Distance (AWCD), and Dispersion Change, together with inferred cell-state dynamics. Accordingly, Equations (1) and (2) should be interpreted as conceptual biological

motivation rather than as components of the computational model. Neither equation is intended to provide a complete quantitative description of tissue dynamics, including changes in total cell abundance resulting from proliferation, apoptosis, or tissue growth.

Supplementary Note S2. Quantitative benchmarking using simulated ground-truth data

To quantitatively assess the reliability of the spatial cell-type abundance estimates underlying the proposed spatiotemporal analyses, we constructed a simulation-based benchmarking framework comparing CARD (1) with five published spatial deconvolution or single-cell mapping methods: RCTD (2), SpatialDecon (3), SPOTlight (4), CellTrek (5), and SpaOTsc (6). Together, these methods represent complementary analytical strategies ranging from spot-level cell-type proportion estimation to single-cell spatial mapping.

Ground-truth simulated data were generated using scDesign3 (7), trained on the UP-8167 annotated parental tumor and corresponding glioblastoma organoid single-cell RNA-sequencing reference containing six cell populations: astrocytes, epithelial cells, neutrophils, OPCs, endothelial cells, and proliferation-like cells. The simulation was parameterized to represent the three disease-related time points used in the study design (TF0, TF1, and TF2). Synthetic single-cell expression profiles were generated at each timepoint and subsequently aggregated into pseudo-spatial spots by mapping them onto the physical spatial coordinates of a real human glioblastoma spatial transcriptomics section. Because the cellular composition of each pseudo-spot was known by construction, the true cell-type proportions were retained as ground truth for every spot and time point.

Each of the six methods was applied to the same simulated single-cell reference and spatial transcriptomic dataset using their default parameter settings. To enable quantitative comparison, outputs were standardized into a common representation containing the time point, spatial spot identifier, spatial coordinates, and predicted cell-type proportions. RCTD (2), SpatialDecon (3), SPOTlight (4), and CARD (1) natively produce spot-level cell-type proportion estimates, whereas CellTrek (5), and SpaOTsc (6) primarily provide single-cell spatial mapping or coupling outputs. For the latter methods, inferred cell-level spatial assignments were aggregated into spot-level cell-type proportions solely for the purpose of this common benchmark.

Method performance was evaluated by comparing predicted and ground-truth cell-type proportions for each cell type and time point using the Pearson correlation coefficient (PCC), root mean square error (RMSE), and mean absolute error (MAE). These metrics were subsequently averaged across cell types and time points. Jensen–Shannon divergence (JSD) was additionally calculated to quantify differences between predicted and ground-truth compositional distributions.

CARD was retained as the deconvolution method for the proposed framework despite not achieving the highest point-estimation accuracy in the benchmark. This selection was based on the combined analytical properties of CARD that are directly relevant to the downstream framework, including spatial correlation modeling of cell-type composition, imputation-based spatial resolution enhancement, and compatibility with single-cell-resolution spatial mapping through CARD_SCMapping (1). Thus, the benchmark was used to quantitatively contextualize the choice of CARD rather than to establish its universal superiority over alternative methods.

This benchmarking analysis evaluates the accuracy of inferred spatial cell-type abundance and provides quantitative assessment of the principal input to the downstream redistribution analyses. It does not constitute independent validation of the downstream spatiotemporal redistribution or inferred cell-state dynamics metrics, nor does it provide evidence of physical cell migration, lineage transitions, or experimentally observed cell-state trajectories.

Supplementary Note S3. Quantitative benchmarking of spatial cell-type abundance estimation

To quantitatively assess the accuracy of the spatial cell-type abundance estimates used as inputs to the proposed spatiotemporal analyses, we benchmarked CARD (1) against five spatial deconvolution or single-cell mapping methods using simulated multi-timepoint data with known ground-truth cell-type proportions. The benchmark included RCTD (2), SpatialDecon (3), CARD (1), SPOTlight (4), CellTrek (5), and SpaOTsc (6), all evaluated using the same simulated single-cell reference and spatial transcriptomic datasets.

Table 2 summarizes the overall agreement between inferred and ground-truth cell-type proportions across cell types and time points. Higher PCC and lower RMSE, MAE, and JSD indicate greater agreement between inferred and ground-truth cell-type proportions.

Supplementary Table S1. Quantitative comparison of benchmarked methods.

Rank	Method	PCC ↑	RMSE ↓	MAE ↓	JSD ↓
1	RCTD (2)	0.7778	0.1225	0.0843	0.0883
2	SpatialDecon (3)	0.7688	0.1301	0.0976	0.1167
3	**CARD** (1)	0.6200	0.1371	0.1037	0.1147
4	SPOTlight (4)	0.1964	0.2073	0.1641	0.2354
5	CellTrek (5)	0.1540	0.2427	0.1776	0.2839
6	SpaOTsc (6)	0.0924	0.2539	0.1720	0.6010

RCTD achieved the highest rank, with a PCC of 0.778, RMSE of 0.123, MAE of 0.084, and JSD of 0.088. SpatialDecon showed comparable performance, with a PCC of 0.769. CARD ranked third among the evaluated methods, with a PCC of 0.620, RMSE of 0.137, MAE of 0.104, and JSD of 0.115. SPOTlight, CellTrek, and SpaOTsc showed lower rank with the simulated ground truth under these proportion-based metrics.

These results indicate that CARD achieved moderate quantitative agreement with the simulated spatial cell-type abundance profiles, although it does not achieve the highest point-estimation accuracy among

the evaluated methods. Importantly, the comparison should be interpreted in the context of the different analytical objectives of the benchmarked approaches.

RCTD (2), SpatialDecon (3), SPOTlight (4), and CARD (1) primarily estimate cell-type proportions, whereas CellTrek (5) and SpaOTsc (6) focus on single-cell spatial mapping or coupling. Their outputs were converted to spot-level proportions for a common benchmark; therefore, their lower performance under proportion-based metrics should not be interpreted as a general assessment of their intended mapping capabilities.

Beyond point-estimation accuracy, the benchmark highlights important methodological differences among the evaluated approaches (Table 3). CARD combines spatially informed cell-type composition modeling with spatial resolution enhancement and the CARD_SCMapping extension for single-cell-resolution spatial assignment (1). In contrast, RCTD (2) and SpatialDecon (3) primarily focus on spot-level proportion deconvolution, whereas CellTrek (5) and SpaOTsc (6) provide complementary approaches for spatial mapping or coupling of individual cells. These differences are relevant because the proposed downstream analyses require both spatially informed abundance estimates and a transition from spot-level composition to an inferred single-cell-resolution spatial representation.

Supplementary Table S2. Summary of analytical capabilities of benchmarked methods

Method	Deconvolution	Single-cell mapping	Resolution enhancement	Colocalization / state analysis	Cell-state dynamics
RCTD (2)	Core	No	No	Indirect (doublet detection)	No
SpatialDecon (3)	Core	No	No	No	No
CARD (1)	Core	Extension	Yes	Indirect	No
SPOTlight (4)	Core	No	No	Partial	No
CellTrek (5)	No	Core	Yes (single-cell)	Yes (SColoc, SCoexp)	Partial (downstream)
SpaOTsc (6)	No	Yes (coupling)	Indirect	Yes (distance/comm.)	Partial (mapping)

Taken together, the benchmark provides quantitative context for the use of CARD as the spatial deconvolution component of the proposed framework, but does not establish CARD as the most

accurate method in terms of point-estimation accuracy alone. Rather, CARD was selected because it provides a combination of spatial correlation modeling, spatial refinement, and compatibility with CARD_SCMapping that is a requirement for the downstream analyses of the proposed framework.

The present benchmark should also be distinguished from direct validation of the downstream spatiotemporal metrics. Because the simulated ground truth specifies cell-type abundance at each spatial location, the benchmark provides a quantitative assessment of the principal spatial input to the redistribution analyses. However, it does not independently establish the accuracy of inferred physical cell movement, lineage transitions, or experimentally observed cell-state trajectories. Direct validation of centroid-based redistribution, dispersion changes, and inferred cell-state dynamics would require additional simulations with explicitly controlled spatial redistribution and cell-state composition changes, ideally complemented by experimentally matched longitudinal, lineage-tracing, or live-imaging data. Such analyses represent important directions for future methodological validation.

Supplementary references

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